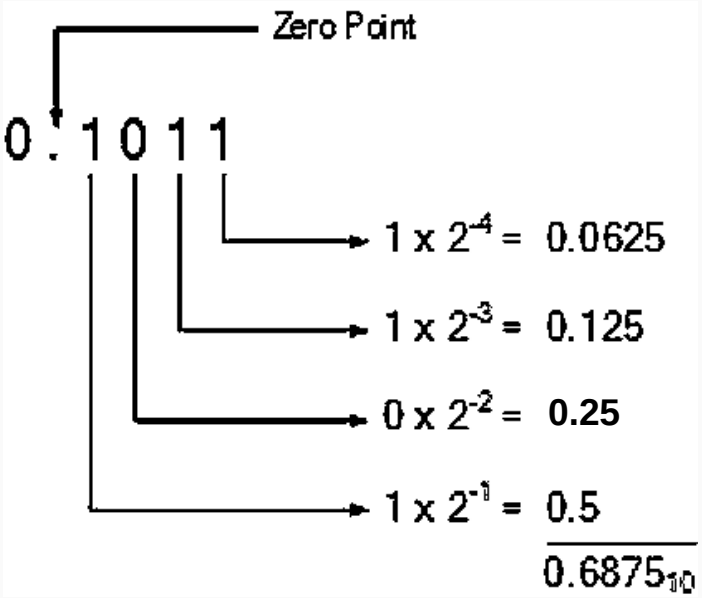
Number System

Fractions Conversion

1. Binary to decimal
    - Multiply each bit (in the fraction) by 2-n, where nis the “weight” of the bit
    - The weight is the position of the bit in the Fraction, starting from (-1)on the left
    - Add the results



- Try to solve :
    - $(0.0111)_2$
    - $(11.0101)_2$
    - $(111.10101)_2$

Let's take an example for n = $(110.101)_2$

Step 1: Conversion of 110 to decimal

- 1102 = (1*22) + (1*21) + (0*20)
- 1102 = 4 + 2 + 0
- 1102 = 6

So equivalent decimal of binary integral is 6.

Step 2: Conversion of .101 to decimal

- 0.1012 = (1* 1/2) + (0* 1/22) + (1*1/23)
- 0.1012 = 1*0.5 + 0*0.25 + 1*0.125
- 0.1012 = 0.625

So equivalent decimal of binary fractional is 0.625

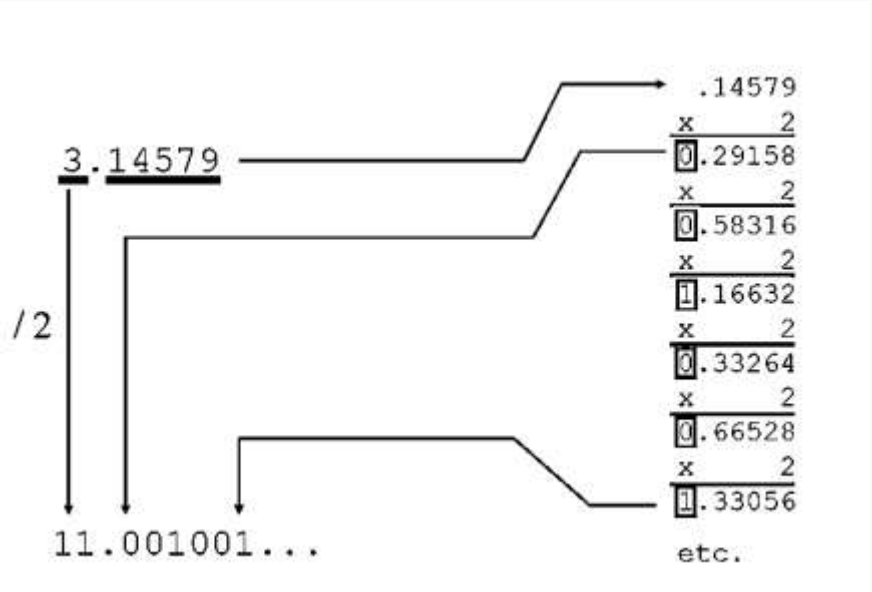
Step 3: Add the result of steps 1 and 2.

- 6 + 0.625 = $(6.625)_{10}$

Fractions Conversion

2. Decimal to Binary

- Decimal Value to Binary(general rule)
- Multiply the fraction by B, and keep track of the remainder
- The first remainder is the leftmost digit in the fraction, ... etc.



- Try to solve :
  - $(2.125)_{10}$
  - $(9.5625)_{10}$
  - $(15.685)_{10}$

Let's take an example for n = $(18.652)_{10}$

Step 1: Conversion of 18 to binary

		18		
16	8	4	2	1
1	0	0	1	0

- $(18)_{10} = (10010)_2$

Step 2: Conversion of .652 to decimal

multiply by 2	Fractional part	Integer Reminder
0.652 x 2	0.25	1
0.25 x 2	0.50	0
0.50 x 2	0	1

- $(18.625)_{10} = (10010.101)_2$

Thanks!